Arael Anaya
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For my final project I plan to study the spring-loaded inverted pendulum (SLIP) model of a hopping/running robot, which is the base for essentially all legged robot models. Despite its initial simplicity (a point mass bouncing on a massless spring leg), the SLIP model captures the center-of-mass dynamics of running in both animals and machines shockingly well. That makes it a great subject for this project: it is simple enough to analyze carefully with the tools from this course, but rich enough to produce very interesting behavior. It also builds naturally from the elastic pendulum system we have analyzed in this classroom, since the stance phase is essentially an inverted spring pendulum.

What makes the SLIP model especially interesting from a mechanics standpoint is that it is a hybrid dynamical system. During the stance phase the body behaves like a spring-loaded pendulum, while in the flight phase it is just a projectile. I will derive the equations of motion for each phase, using the Lagrangian formulation in polar coordinates for the stance phase, and work out the switching conditions at touchdown and liftoff that connect the two phases.

Rather than just analyzing the stability at a single equilibrium point, I will study the stability of the hopping gait itself. From quick research, the natural tool for this is an apex return map (a Poincaré map in which a periodic gait appears as a fixed point whose stability can be evaluated numerically). I will implement this model in Python, using event-based numerical integration, and run a parameter study on quantities like spring stiffness and touchdown angle to map out which combinations produce stable hopping. There is substantive literature on SLIP dynamics, so there is plenty to draw from for the paper and video presentation.